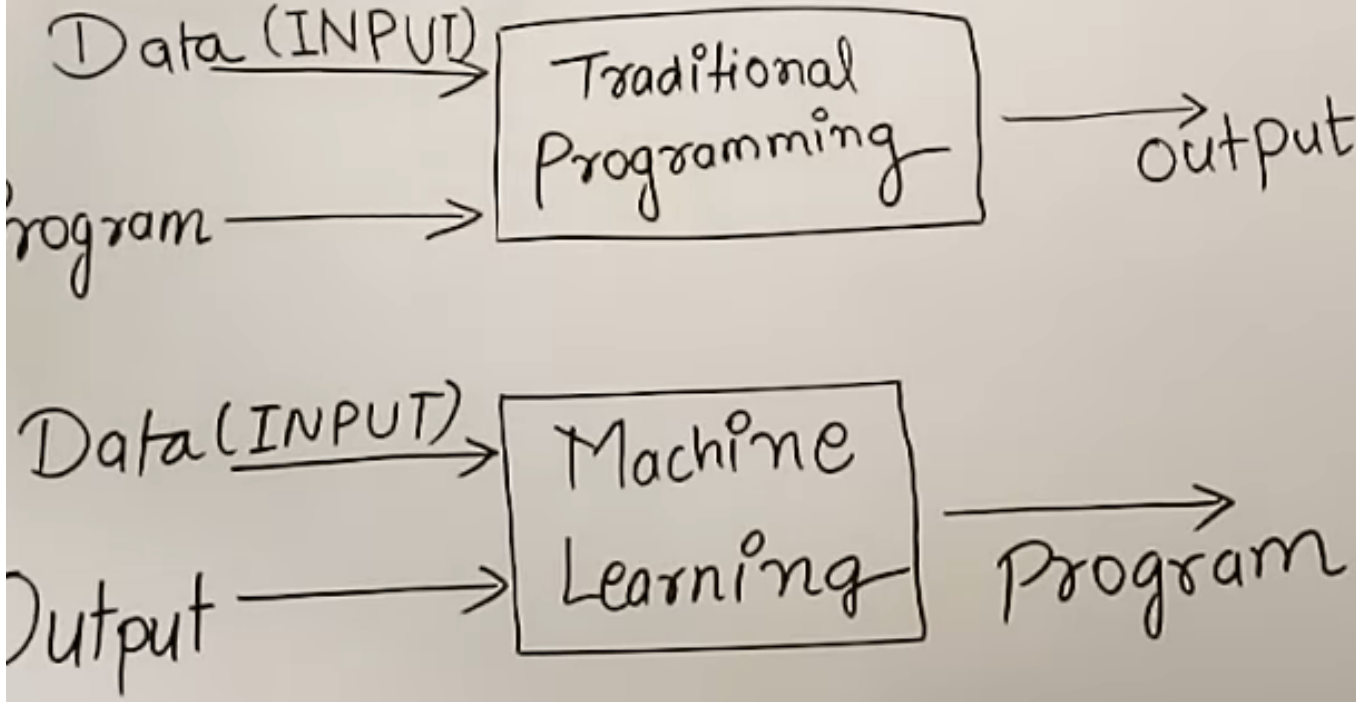
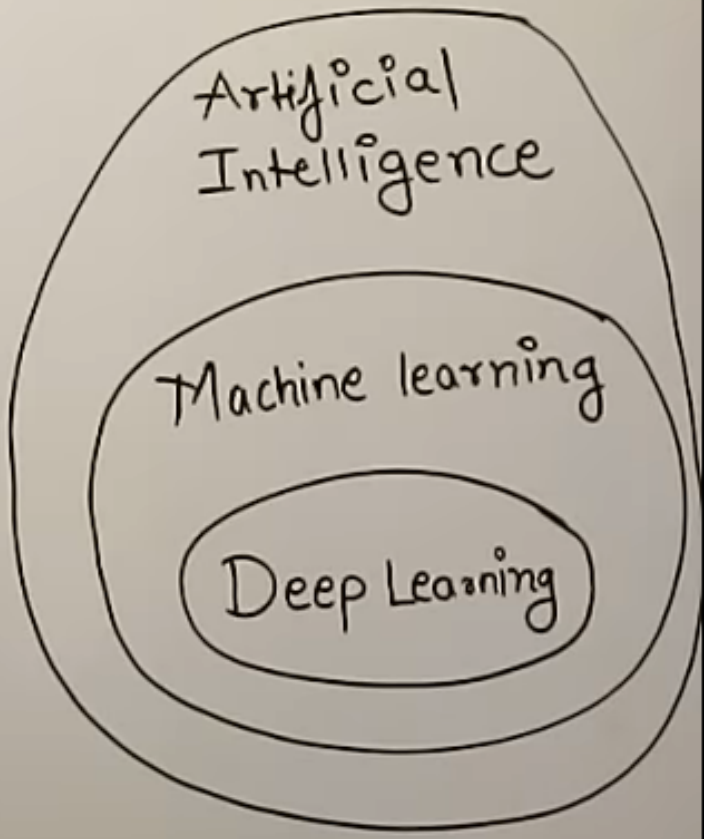
Machine learning

- It is the field of study that gives Computers the Capability to learn without being explicitly programmed.





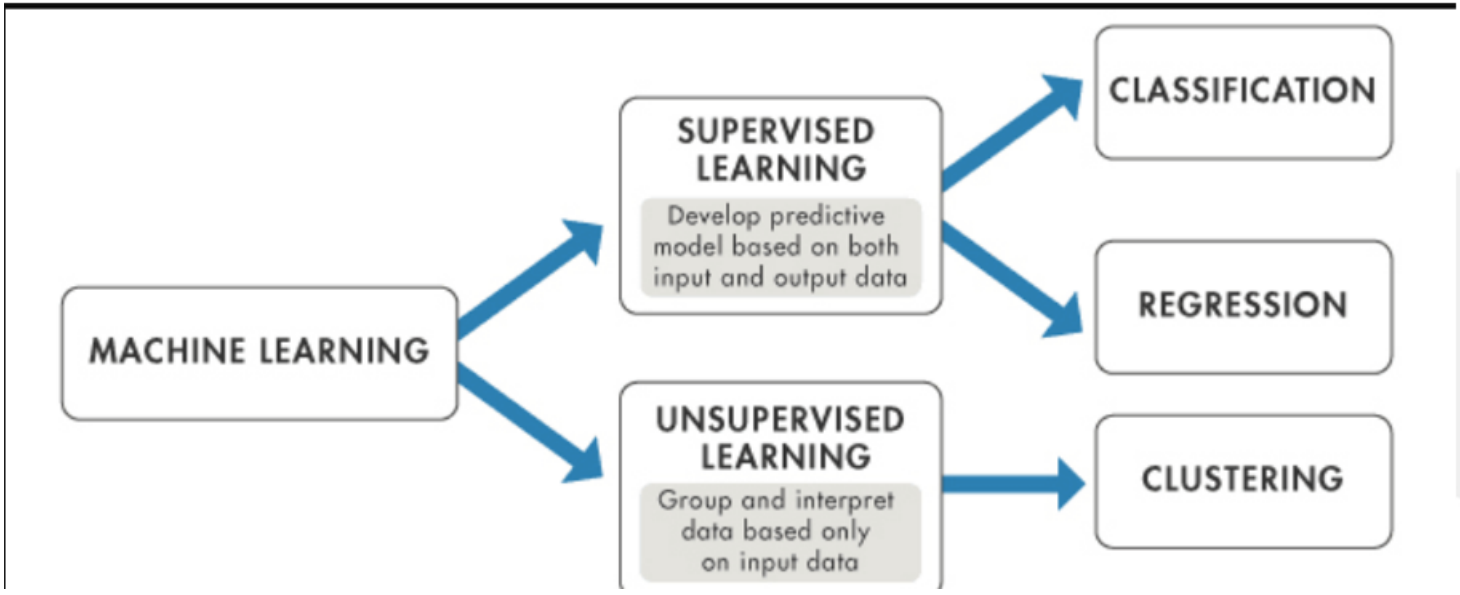
Definitions

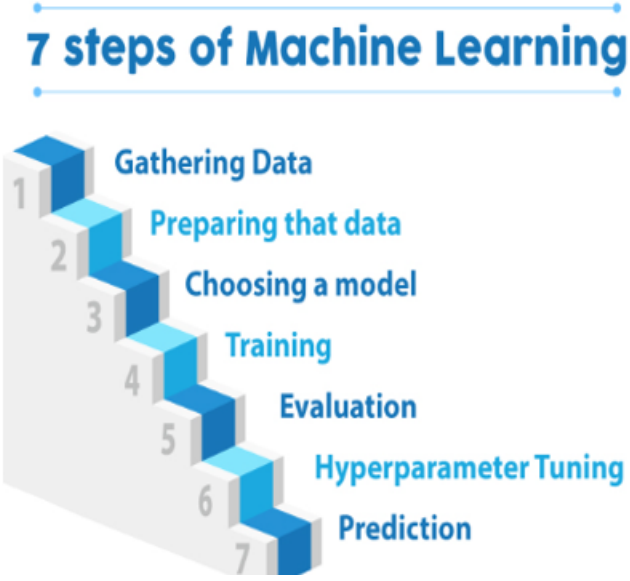
Artificial intelligence (AI): A broad discipline with the goal of creating intelligent machines, as opposed to the natural intelligence that is demonstrated by humans and animals. It has become a somewhat catch all term that nonetheless captures the long term ambition of the field to build machines that emulate and then exceed the full range of human cognition.

Machine learning (ML): A subset of AI that often uses statistical techniques to give machines the ability to "learn" from data without being explicitly given the instructions for how to do so. This process is known as "training" a "model" using a learning "algorithm" that progressively improves model performance on a specific task.

Reinforcement learning (RL): An area of ML concerned with developing software agents that learn goal-oriented behavior by trial and error in an environment that provides rewards or penalties in response to the agent's actions (called a "policy") towards achieving that goal.

Deep learning (DL): An area of ML that attempts to mimic the activity in layers of neurons in the brain to learn how to recognise complex patterns in data. The "deep" in deep learning refers to the large number of layers of neurons in contemporary ML models that help to learn rich representations of data to achieve better performance gains.



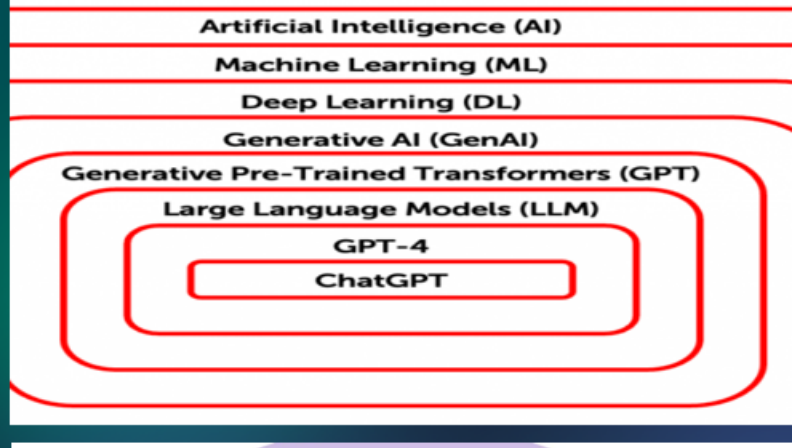


What are Machine learning algorithms?

Machine learning algorithms are the engines of machine learning, meaning it is the **algorithms that turn a data set into a model.**



AI Terminology



A model for thinking about AI, ML, DL, and generative AI



Artificial intelligence (AI)

Any technique that allows computers to mimic human intelligence using logic, if-then statements, and machine learning



Machine learning (ML)

A subset of AI that uses machines to search for patterns in data to build logic models automatically



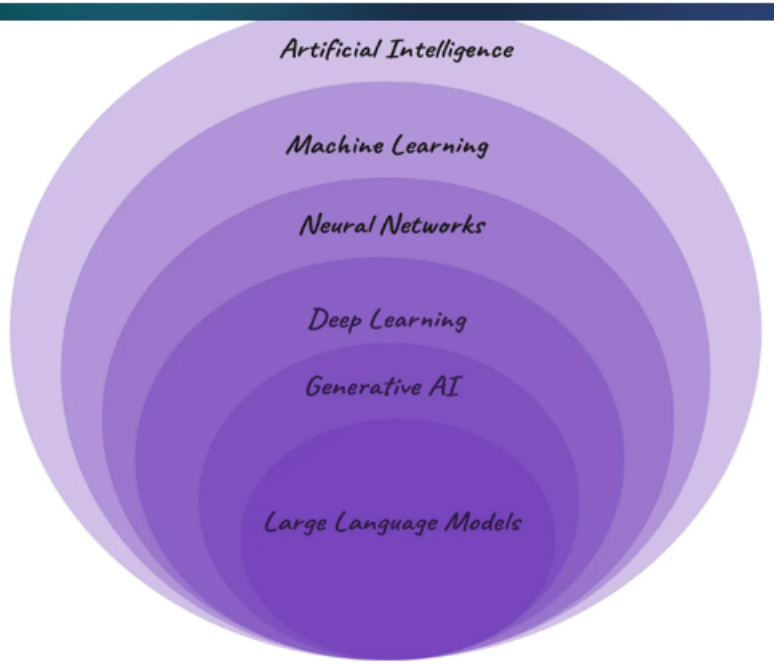
Deep learning (DL)

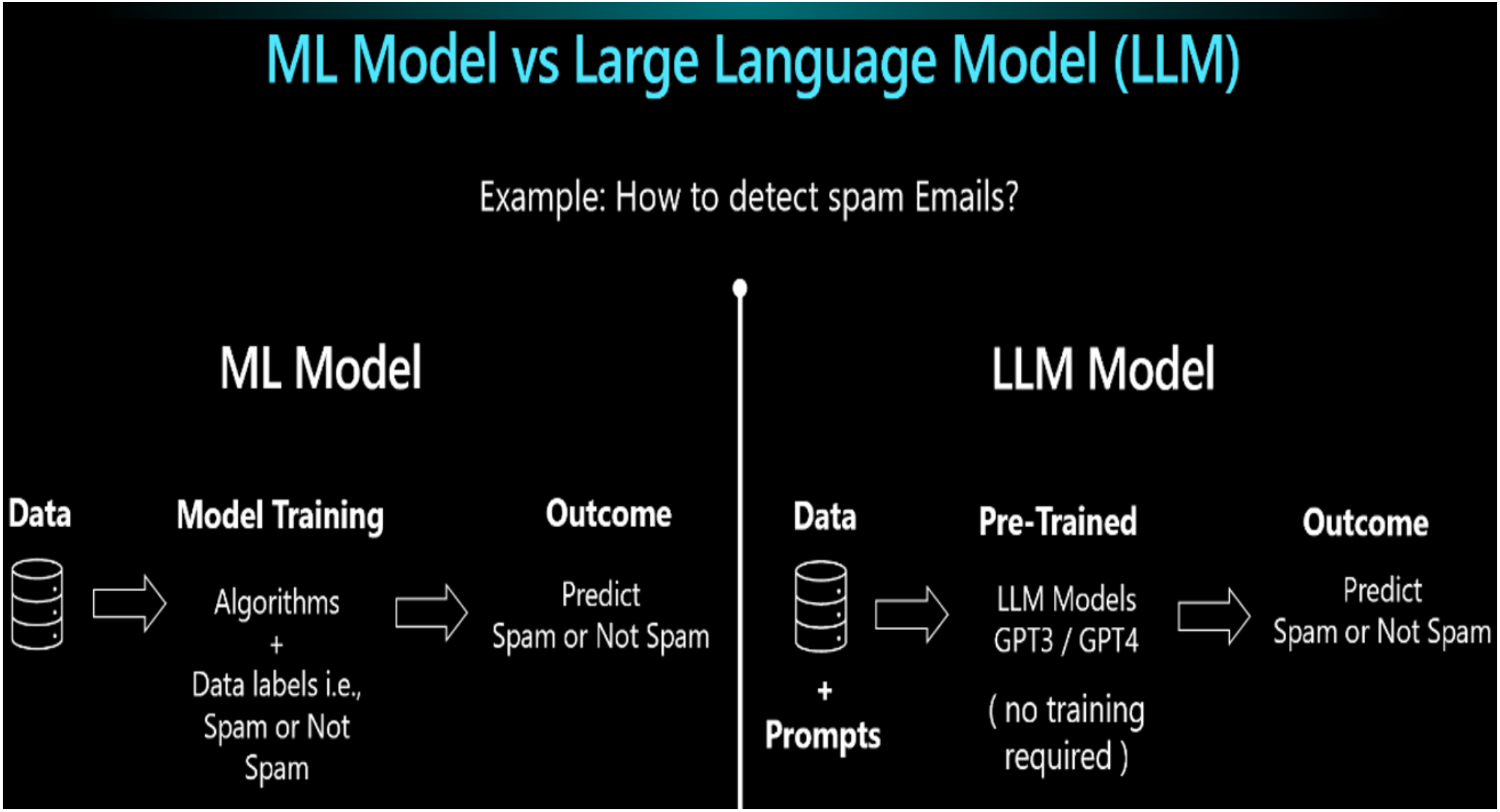
A subset of ML composed of deeply multi-layered neural networks that perform tasks like speech and image recognition



Generative AI

Powered by large models that are pretrained on vast corpora of data and commonly referred to as foundation models (FMs)




ML Model vs Large Language Model (LLM)
Example: How to detect spam Emails?
ML Model
Data
Model Training
Algorithms
+
Data labels i.e.,
Spam or Not
Spam
Outcome
Predict
Spam or Not Spam
LLM Model
Data
+
Prompts
Pre-Trained
LLM Models
GPT3 / GPT4
(no training
required)
Outcome
Predict
Spam or Not Spam